Basic Principles of Medicine 1
Module : Foundations to Medicine
MSK Lecture No: # 22

Lecture Title: Inherited Disorders of the Musculoskeletal System

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Directed learning activity to prepare:

1. **The two paper-case readings which are posted as PDF**
   - Osteogenesis imperfecta (OI)
   - Duchenne muscular dystrophy (DMD)
1. **The videos for:**
   - OI
   - DMD
3. **Recommended Readings from the textbook**
   - Page 33 for Clinical Snapshot 2.2; Charcot-Marie-Tooth disease
   - Page 46-47 for Marfan syndrome; also Figure 3.11
   - Page 46-47, 48 for OI (also Fig 3.12, on p48)
   - Page 47 for Achondroplasia; also Fig 3.13, on p48
   - P 92-94 and 219 for Alzheimer disease (on p219, (see top left-hand column).
   - P 171-180 for Neurofibromatosis - Note: focus on *NF1*.
4. **Go to the website: http://www.korfgenetics.com/factsheets.asp** (these are 1-page synopses)

   In Neuromuscular disorders select
   - 41. Duchenne/Becker muscular dystrophy

   In Cancer predispositon syndromes select
   - 18. Neurofibromatosis

   In Skeletal/connective tissue select:
   - 64. Osteogenesis imperfecta
   - 65. Marfan syndrome
   - 67. Achondroplasia

Pre-Lecture - Required Reading and Objectives

DLA Title: Genetics Cases OI and DMD

SOM.MKIII.BPM1.2.MSK.1.GNET.0154 Review the genetic mechanisms in DMD, OI, NF-1, CMT, Marfan Syndrome, Achondroplasia and Alzheimer's disease.

SOM.MKIII.BPM1.2.MSK.1.GNET.0154	Review the genetic mechanisms in DMD, OI, NF-1, CMT, Marfan Syndrome, Achondroplasia and Alzheimer's disease

Note:

The genetics portion of MSK module consists of 4 total sessions:

- 2 DLAs
- 2 Flipped Lectures

Objectives: DLA and Lecture

SOM.MK.III.BPM1.2.MSK.1.GNET.0156	Discuss elevated creatine kinase levels in DMD and BMD patients
SOM.MK.III.BPM1.2.MSK.1.GNET.0157	Describe the inheritance pattern of Duchenne muscular dystrophy and discuss the clinical features
SOM.MK.III.BPM1.2.MSK.1.GNET.0158	Compare and contrast the allelic heterogeneity in the DMD gene that leads to either Becker muscular dystrophy or Duchenne muscular dystrophy
SOM.MK.III.BPM1.2.MSK.1.GNET.0159	Analyze the DMD gene and discuss why it is sometimes a hotspot for mutation in the human genome
SOM.MK.III.BPM1.2.MSK.1.GNET.0160	Discuss de novo mutations which lead to 1/3 of all cases of DMD
SOM.MK.III.BPM1.2.MSK.1.GNET.0161	Explain the effects of skewed X-inactivation leading to a manifesting heterozygote in females
SOM.MK.III.BPM1.2.MSK.1.GNET.0162	Describe the structure of collagen
SOM.MK.III.BPM1.2.MSK.1.GNET.0163	Describe the general features of osteogenesis imperfecta
SOM.MK.III.BPM1.2.MSK.1.GNET.0164	Describe how classic non-deforming OI with blue sclera (formerly known as OI type 1) using the haploinsufficiency model.
SOM.MK.III.BPM1.2.MSK.1.GNET.0165	Describe the perinatally lethal osteogenesis imperfecta (formerly known as OI type 2) using the dominant negative model.
SOM.MK.III.BPM1.2.MSK.1.GNET.0154	Review the genetic mechanisms in DMD, OI, NF-1 , CMT, Marfan Syndrome, Achondroplasia and Alzheimer disease

Clicker Questions Next!

Please prepare to answer
the clicker questions
on the slides.

A 17-year-old boy is brought to the physician because of a 3-year history of weakness in his feet that has progressed to foot drop and difficulty with walking (his toes scrape the ground when he walks). He has depressed knee and ankle jerk reflexes. He has very high arches and must wear special shoes. His father and his maternal grandmother are similarly affected. Genetic test shows a pathogenic duplication of a gene on chromosome 17. Which of the following disorders does this patient most likely have?

A. Marfan syndrome 0%
B. Neurofibromatosis 0%
C. Becker muscular dystrophy 0%
D. Charcot-Marie-Tooth disease 0%
E. Friedreich ataxia 0%
F. Myotonic dystrophy 0%

See clinical snapshot 2.2 (p33) in Korf and Irons; 4th edition

Genetic principle: deletions are usually more deleterious than duplications

- If duplicated, an area of chromosome 17 will cause Charcot-Marie-Tooth disease (most common cause of CMT)
  - If deleted, this same area will cause a more severe genetic disorder (we won’t describe it here).
- Parallel example:
  - Trisomy 21 is a cause of Down syndrome
  - Monosomy 21 would be embryonic lethal
- Principle: gain of genetic information is usually better tolerated than loss of genetic information

A 15-year-old girl is brought to the physician by her parents because of a 10-year history of steady accumulation of a multitude of neurofibromas. Targeted gene analysis from peripheral blood cells shows germline inheritance of a pathogenic *NF1* mutation. Which of the following molecular findings would be expected upon analysis of a biopsy from one of the neurofibromas from this patient?

A. Increased *NF1* mRNA	0%
B. Gain of *NF1* function	0%
C. Anticipation in the other *NF1* gene	0%
D. Increased RAS activity	0%
E. Increased apoptosis	0%
F. Activation of a cell cycle checkpoint	0%

Must understand cancer predisposition syndromes: Concept of a tumor suppressor gene Vs. concept of protooncogene.

See Figure 10-4 in the Korf and Irons textbook 4th edition

Review the function of RAS

- What is a RAS-GAP?
- Which pathway is regulated by RAS?

As a RAS-GAP, neurofibromin is a tumor suppressor; NF1 is a tumor suppressor gene

A 19-year-old man comes to the physician because of a lifetime history of steady accumulation of neurofibromas. Targeted gene analysis from peripheral blood cells shows germline inheritance of a pathogenic *NF1* mutation. Which of the following genetic events promoted the tumor formation in this patient?

A. Increased amounts of *NF1* mRNA 0%

B. Activation of the G1/S checkpoint 0%

C. Anticipation in the other *NF1* gene 0%

D. Stimulation of the *TP53* gene 0%

E. Second hit in the other *NF1* gene 0%

See chapter 10 in the Korf and Irons textbook 4th edition

Review the concept of…

Loss of heterozygosity as an explanation for the autosomal dominant inheritance of some cancer predisposition syndromes:

- NF1 - neurofibromatosis
- BRCA1/2 Hereditary Breast and Ovarian Cancer syndrome
- RB – retinoblastoma
- APC – Familial adenomatous colon cancer
- MSH2 (mismatch repair pathway) – Lynch syndrome

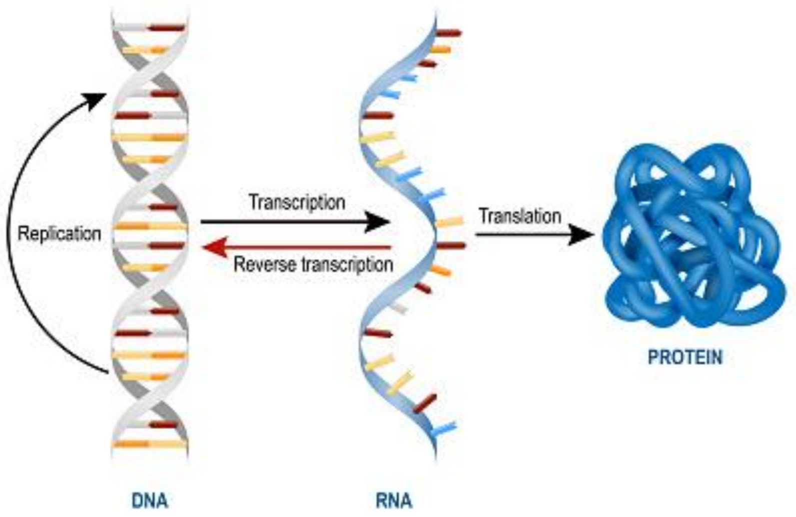
  - Compare and contrast mutations that promote cancer progression:
    - Tumor suppressor genes VS. proto-oncogenes / oncogenes

A 25-year-old woman comes to the physician for follow-up examination after being diagnosed with neurofibromatosis. She has multitude of neurofibromas and Lisch nodules on the iris of her eyes. Targeted gene analysis from peripheral blood cells shows germline inheritance of a pathogenic *NF1* variant that is suspected to interfere with normal splicing. Which of the following would most likely be used to verify that this mutation actually affects a splice site?

A.	Create and sequence cDNA from *NF1* mRNA	0%
B.	Obtain a G-band karyotype of the patient	0%
C.	Use the SNP chip	0%
D.	Look for microsatellite repeats	0%
E.	Exome sequence by next generation sequencing (NGS)	0%

See Figure 10-3 in the Korf and Irons textbook 4th edition

Review Central Dogma of Cell Biology



- DNA directs synthesis and orientation of RNA
- RNA directs synthesis and orientation of protein
- DNA can be replicated
- Rule breakers exist:
  - Consider reverse transcriptase

A 25-year-old man with NF comes to the physician for follow-up examination. Targeted analysis of DNA from peripheral blood cells shows inheritance of a germline pathogenic variant mutation in *NF1*. Biopsy of 100 different neurofibromas shows that a different mutation occurred as the 'second hit' in each of the benign tumors. Which of the following is the best explanation for this observation?

A. Triplet repeat expansion — 0%
B. The gene is in a heterochromatin region — 0%
C. Many SNPs are in the locus — 0%
D. NF1 is a mutation hotspot — 0%
E. Enhancer sequences are found 10 Kb upstream — 0%

See Figure 10-3 in the Korf and Irons textbook 4th edition

NF1 is in the top 10 of the biggest genes in the human genome

- *DMD* is also one of the biggest genes
- There are a lot of places in the gene that can be mutated
- Missense, nonsense, splice site, deletions, anything ...
- Analysis of the *NF1* gene is therefore quite difficult

- There are many different ways to cause loss of function of a gene
- So, there are many different possible mutations that can cause loss of function

Some percentage of individuals thought to have NF1 never get molecular confirmation

- Either the mutation can’t be found
- Analysis of *NF1* gene is difficult

- or they have NF due to mutation of some other gene
- Locus heterogeneity

A couple who have a child with classic non-deforming OI with blue sclera (Type I) come to the physician for advice about recurrence risk as they are expecting a second child. The *COL1A1* gene in their affected child has been sequenced, and a pathogenic mutation has been identified. What is the next step that needs to be taken to address the question of recurrence risk?

A.	Test for appropriate splicing of the pre-mRNA	0%
B.	Check for the mutation in both parents	0%
C.	Test biochemical properties of purified protein	0%
D.	Check other structural genes such as fibrillin	0%
E.	Determine if the mutation in the child displays a dominant negative inheritance pattern	0%

Read through the chapter 10 case in the Korf and Irons 4th edition textbook for a complete discussion of the importance of appropriate data collections and analysis with respect to inheritance

A 3-year-old boy who has had many broken bones is brought to the physician by his parents. He is diagnosed with OI. Genetic tests show that he has a missense mutation in *COL1A1* causing a glycine to a cysteine substitution in an exon near the 3' end of the gene. Which of the following is most likely the genetic mechanism explaining the pathology in this patient?

A. Haploinsufficiency 0%
B. Dominant negative 0%
C. Autosomal recessive 0%
D. Loss of heterozygosity 0%
E. Gain of function 0%
F. Polygenic effect 0%
G. Incomplete penetrance 0%

Point mutation that causes dominant negative effect. Depending on the mutation, severity might be variable. There are many of these various other subtype forms of OI (not discussed in this module).

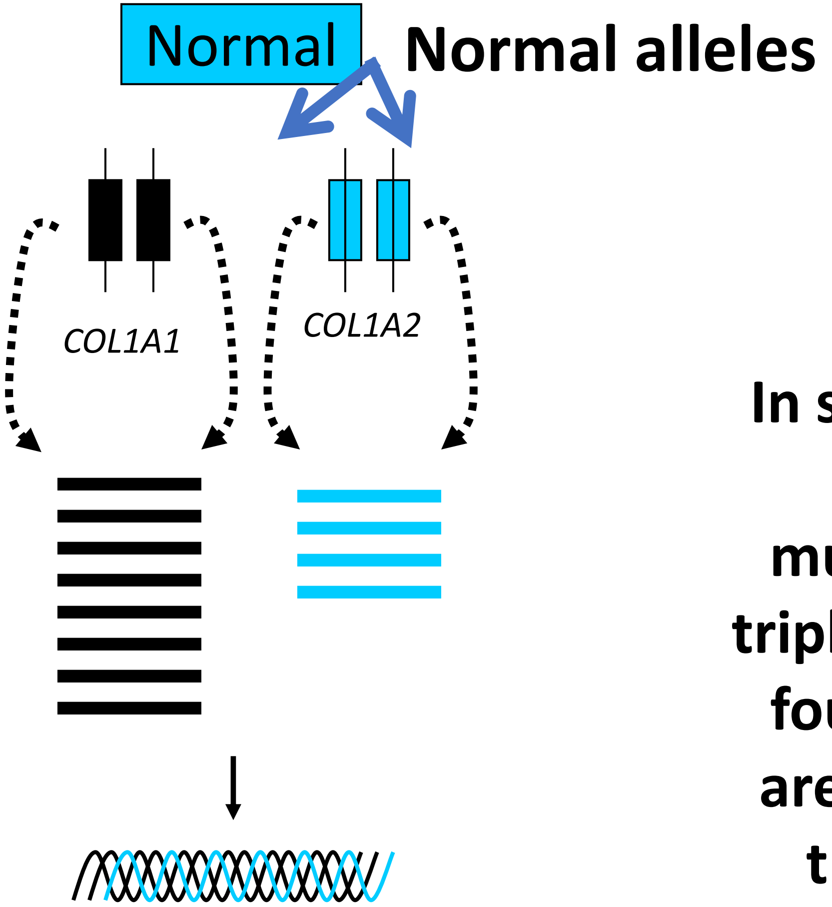
Collagen Gene Mutations

Point mutation that causes dominant negative effect. Can be very severe type II OI, with perinatal lethality

Loss of expression of one allele – haploinsufficiency mechanism of OI - this is type I OI with blue sclera

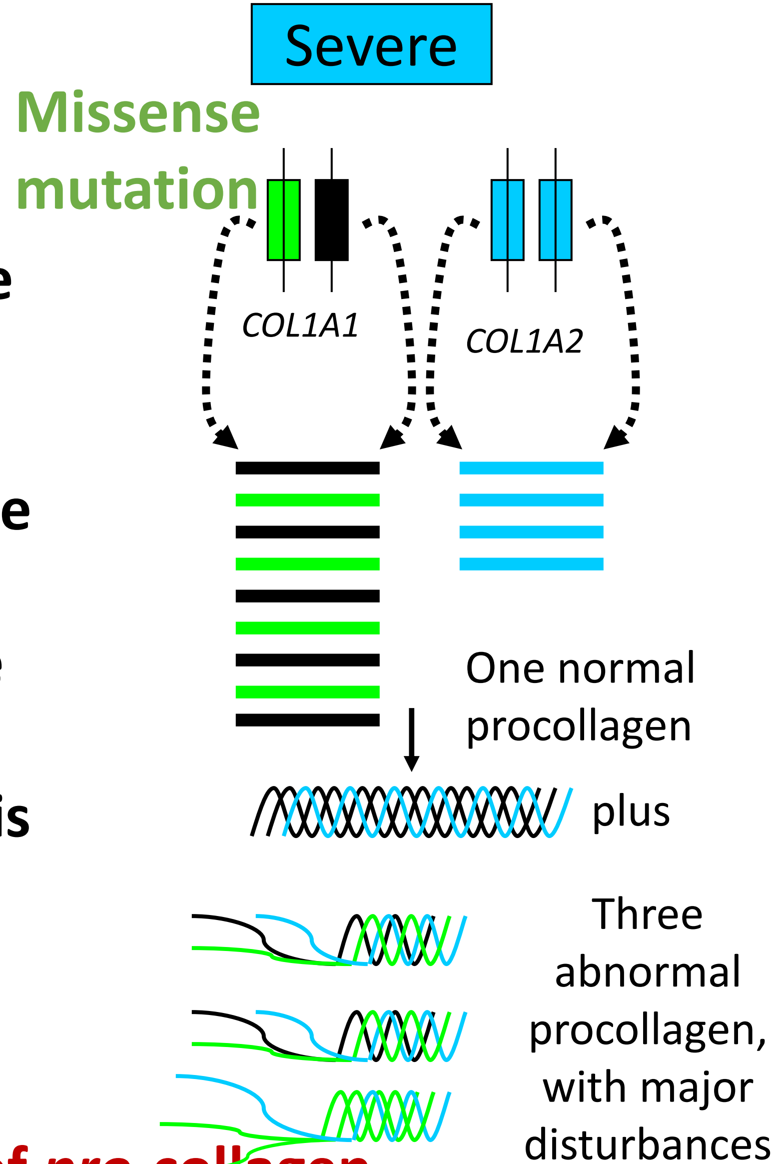
Functional variants of *COL1A1* / *COL1A2* that produce collagen that is "*good enough*". Person does not have OI

Dominant negative Loss of function OI - severe



Four normal procollagen molecules with three chains each (triple helix)

In severe OI one copy of the gene has a missense mutation that disrupts the triple helix. Then three of the four collagen triple helixes are disrupted and the bone that is formed is greatly weakened (and worse, in this case, only 25% of the collagen is normal).



The pathogenic missense mutation leads to formation of *pro*-collagen that after assembly forms collagen that is weak, so bones break easily

The dominant negative form of osteogenesis imperfecta

Depending on how *BAD* (or how severe) *the* effect of the pathogenic missense mutation in *COL1A1* is:

Might be: perinatally lethal OI (Type 2)

or

Other forms of OI that allow the patient to survive; but with severely weakened bones

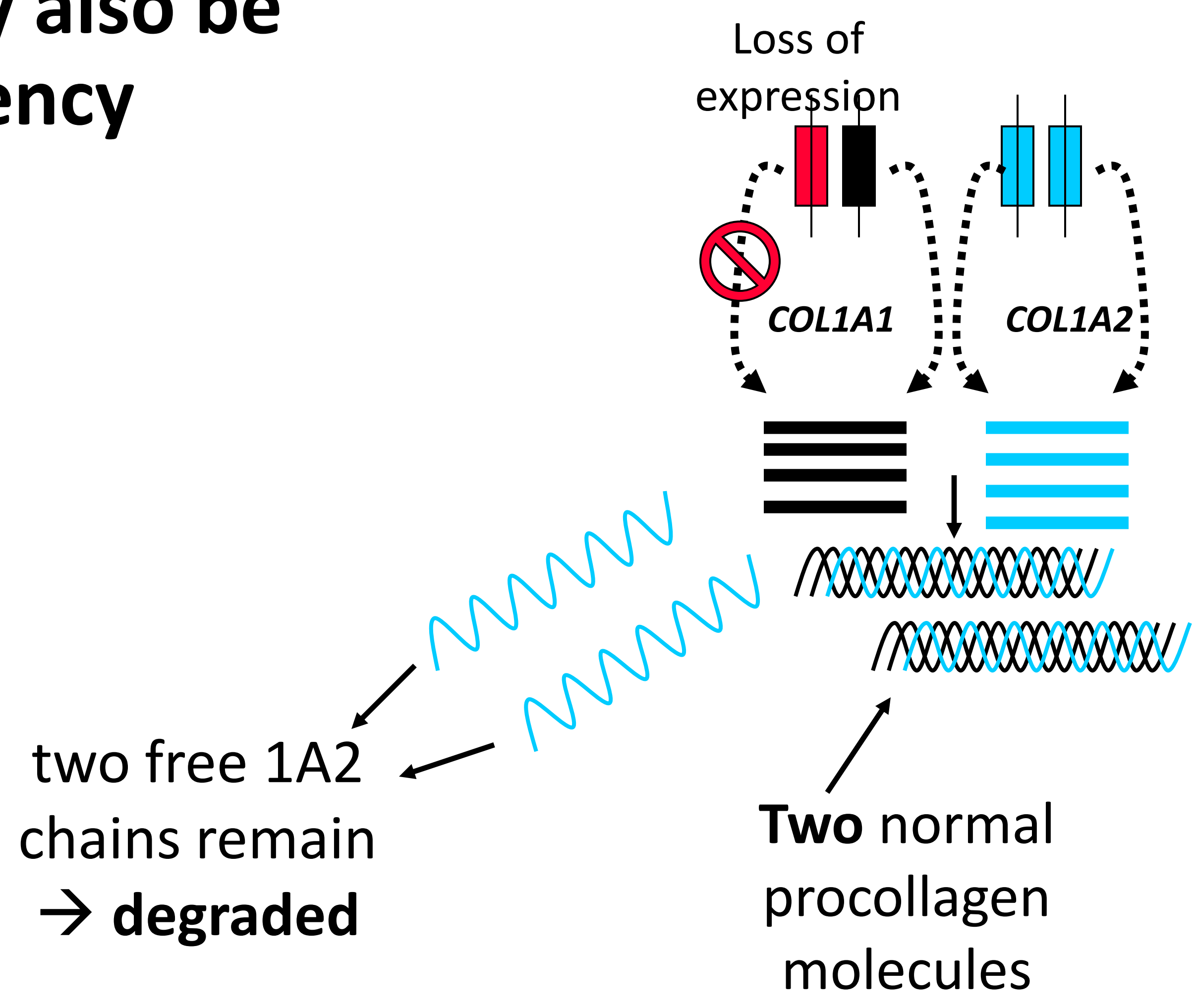
(*Such as the person described in the previous question*)

Osteogenesis imperfecta may also be caused by haploinsufficiency

The collagen molecule is a triple helix

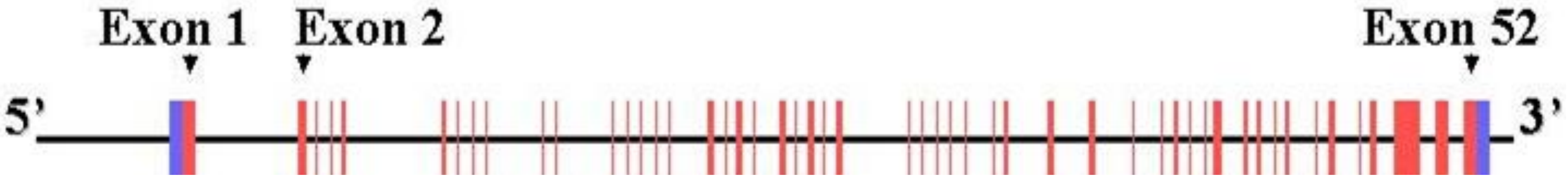
In mild OI one copy of the gene is knocked out, half the amount of collagen 1A1 is produced and less triple helix collagen can form

This is Classic Non-deforming OI with blue sclera



Loss of expression of 1 copy of *COL1A1* leads to <u>less</u> collagen being formed (OI), but the collagen that is formed, is normal, and strong. The problem is that there isn't enough of it. Think of some genetic mutations that would lead to loss of expression:

Genetic analysis was done postmortem for a baby who died soon after delivery. This testing shows a mutation in the *COL1A1* gene, suggesting defective collagen synthesis. Which of the following variants is most likely responsible for this pathology?



A. Missense mutation in exon 45 — 0%
B. Nonsense mutation in exon 10 — 0%
C. Mutation in the promoter (abolishes expression) — 0%
D. Frameshift due to splice site mutation in exon 5 — 0%

:57

Compare and contrast the effect of a mutation in an enzyme (or something with an activity, like CFTR) Vs. a protein such as collagen which assembles to form higher level structures

Enzyme, or channel:

- Missense mutation might lead to a protein that has residual activity
  - If enough residual activity is retained, effect could be sub-clinical
- Nonsense (early stop codon) mutation likely to cause no protein to be made at all
  - Complete loss of functional protein (no residual activity

A protein like collagen:

- Missense mutation might lead to a formation of a protein which disturbs the higher-level structure
  - Dominant negative
- Nonsense mutation will cause less structures to be made
  - But what is made will be normal

Disorders that affect connective tissue, muscle and skeleton: concept map

(Summary map for students)

Connective tissue, bones and muscle

Achondroplasia
- Disproportionate short stature
- Due to unregulated tyrosine kinase signaling
- mutation hot spot
- gain of function

Marfan syndrome
- Due to defective fibrillin in connective tissue
- Autosomal dominant
- Could be Haploinsufficiency or dominant negative (dom neg is usually more severe)

DMD and BMD
- X-linked
- Muscle fiber integrity is lost, or reduced
- Genotype phenotype relationship well understood (DMD vs. BMD)

Ehlers-Danlos syndrome
- Due to defective collagen in the extracellular matrix
- Collagen in the connective tissue
- ***Very*** clinically heterogeneous
- Most forms are AD could be either haploinsufficiency, or dominant negative
- Some forms are more lethal than others
- More heterogeneous than OI

Osteogenesis imperfecta
- Due to defective collagen in the bones (COL1A1/COL1A2)
- Either not enough collagen (haploinsufficiency; type 1)
- Or most/all collagen formed is defective (dominant negative)
- Also, clinically heterogeneous
- Most forms are AD, some rare forms might have other inheritance patterns (we don't discuss these)

Homocystinuria
- Individuals might have some of the features of "classic" Marfan syndrome
- In 1960 homocystinuria was thought to be a subtype of Marfan syndrome
- This was sorted out after elevated levels of homocysteine were detected, and found to interact with connective tissue

Two patients with muscular dystrophy are compared. Both have a deletion of similar size ~100 kb in the 5’ area of the *DMD* gene. Patient one has mild Becker symptoms, and patient 2 has severe Duchenne symptoms. This is because patient 2 has a

A. Synonymous mutation 0%
B. Frameshift mutation 0%
C. Insertion of one codon 0%
D. Transversion mutation 0%
E. Transition mutation 0%
F. Silent mutation 0%

:72

Reproductive capability is being discussed in patients who have CF, NF1, DMD and BMD. Which is the best example of this genetic principle?

A. Men with DMD have a genetic fitness of 1 since most males with the disorder may not reproduce — 0%

B. Men with NF1 are unable to reproduce — 0%

C. Genetic fitness of female carriers with DMD is close to zero since they are usually mildly affected — 0%

D. Men are more likely to pass on a BMD disease allele to daughters than a DMD disease allele — 0%

Men with CF do not have a vas deferens (CAVD), so they are unlikely to be able to reproduce = low genetic fitness

Self Study: 50 – 60% of all women who are carriers of a variant that causes DMD in boys have cardiac abnormalities at some point in their lives.

Think about this, and discuss why each of the following might increase the risk of cardiac complications in women who are carriers of a DMD causing variant:

1. The woman also has Turner syndrome
2. The woman also is a heavy smoker
3. The woman also has skewed X-inactivation
4. Her father has Becker muscular dystrophy, and her mother is a carrier of a mutation that would cause Duchenne muscular dystrophy in a boy. She inherits a pathogenic variant from both parents.

A pediatric patient with DMD typically has such poor health that he is unable to reproduce, so we say that the genetic fitness of this patient is near zero (0). Since this patient is unable to pass on his mutation, what keeps the disease incidence rate relatively stable in the population?

A. BMD alleles may progress to DMD alleles 0%
B. Non-disjunction in older women affects DMD gene 0%
C. Sperm of older men have a higher mutation load 0%
D. 33% of DMD cases are due to *de-novo* mutation 0%
E. Increased risk of non-disjunction in women >35 yo 0%

Mutation rate in an X-linked disorder that has a genetic fitness of 0 in boys

- 33% of cases must be due to new mutation if the frequency of the disorder is not changing in the population
- This is because any boy who has the mutation may not reproduce, and hence, that mutation is lost from the population
- There are three places where an X chromosome may be found in humans
  - Females have 2 X chromosomes (this is two places)
  - Males have 1 X chromosome (this is one place)
  - 2 + 1 = 3

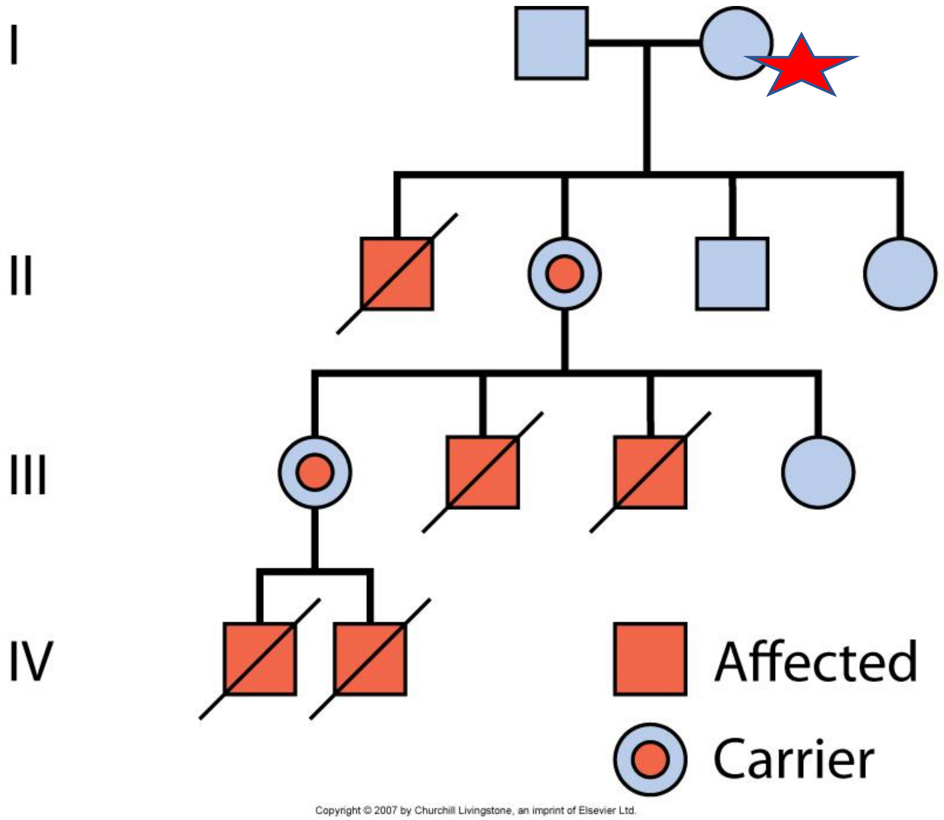
- From the DLA… Think about some reasons why the *DMD* gene exhibits such a high rate of mutation.

Gene name: dystrophin
Gene symbol: *DMD*

Example of X-Linked Recessive: Duchenne Muscular Dystrophy

2/3 of cases transmit down lineages from carrier mothers

1/3 of cases are caused by new mutations (*de novo*) where there has been no previous family history of the disease.



Family tree DMD being transmitted by carrier females:

Affects males, who typically do not reproduce (low genetic fitness)

★ I-2 is likely to be a carrier since she had an affected child, and she had a carrier daughter. The mutation that she carries was either passed to her by (she may also be germline mosaic, but this is very rare compared to *de novo* mutation)

Duchenne Muscular dystrophy tends to be lethal before the age of 30 (Males die), but due to the pathology, very few males are able to reproduce = low genetic fitness

The end

Unless there is more time, then we can do some more questions as seen on the next 4 slides